1. The particular (separable) solutions of the Laplace's equation $\nabla^2 T = 0$, which obey the homogeneous boundary conditions at $x=0$, $x=a$ and $y=0$, are the following:

$$T_p(x,y) = B_k \sinh\frac{\pi k y}{a} \sin\frac{\pi k x}{a},$$

$$k = 1, 2, 3, \ldots$$

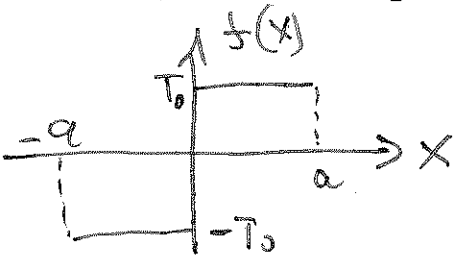
Then the solution can be written as

$$T(x,y) = \sum_{k=1}^{\infty} B_k \sinh\frac{\pi k y}{a} \sin\frac{\pi k x}{a}$$

At $y=a$ we demand

$$T(x,a) = \sum_{k=1}^{\infty} B_k \sinh(\pi k) \sin\frac{\pi k x}{a} = T_0.$$

To find B_k, we need to expand T_0 in the Fourier series of sines:



Let $B_k \sinh(\pi k) = b_k$

$$b_k = \frac{2}{2a}\int_{-a}^{a} f(x) \sin\frac{\pi k x}{a}\,dx = \frac{2T_0}{\pi k}\left[1-(-1)^k\right]$$

Result: $T(x,y) = \sum_{n=0}^{\infty} C_n \sinh\frac{2n+1}{a}\pi y \sin\frac{2n+1}{a}\pi x,$

where $C_n = \dfrac{4T_0}{\pi(2n+1)\sinh[\pi(2n+1)]}$.

2. The solution must be of the form

$$u(r,\varphi) = a_0 + b_0 \ln r + \sum_{n=1}^{\infty} \left[\left(a_n r^n + b_n r^{-n} \right) \cos(n\theta) + \left(c_n r^n + d_n r^{-n} \right) \sin(n\theta) \right]$$

Rewriting the boundary conditions as

$$u\big|_{r=1} = \frac{3}{2} + \frac{1}{2}\cos 2\varphi, \quad u\big|_{r=2} = \frac{1}{2} - \frac{1}{2}\cos 2\varphi,$$

we obtain

$$u(r,\varphi) = \frac{3}{2} - \frac{\ln r}{\ln 2} + \left(\frac{2}{3r^2} - \frac{r^2}{6} \right) \cos 2\varphi.$$

3. We go over to the characteristics

$$\begin{cases} \xi = x + t \\ \eta = x - t \end{cases}$$

and obtain $\dfrac{\partial^2 u}{\partial \xi \partial \eta} + \dfrac{3}{2} = 0.$

The general solution is

$$u(\xi,\eta) = -\frac{3}{2}\xi\eta + f(\xi) + g(\eta), \text{ where}$$

f and g are arbitrary functions.

$$\Rightarrow u(x,t) = -\frac{3}{2}(x^2 - t^2) + f(x+t) + g(x-t).$$

$$\frac{\partial u(x,t)}{\partial t} = 3t + f'(x+t) - g'(x-t).$$

The initial conditions yield:

$$\begin{cases} -\frac{3}{2}x^2 + f(x) + g(x) = x^2 & (1) \\ f'(x) - g'(x) = 4x & (2) \end{cases}$$

$$(1) \Rightarrow \begin{cases} f(x) + g(x) = \frac{5}{2}x^2, \\ (2) \Rightarrow \quad f(x) - g(x) = 2x^2 + C, \end{cases}$$

and we find $f(x) = \frac{9}{4}x^2 + \frac{C}{2}$ and

$$g(x) = \frac{1}{4}x^2 - \frac{C}{2}.$$

Therefore, $u(x,t) = -\frac{3}{2}(x-t)^2 + \frac{9}{4}(x+t)^2 + \frac{(x-t)^2}{4} =$

$$= x^2 + 4xt + 4t^2 = (x+2t)^2.$$

4. We first find the Green's function from eqn. $\frac{d^2G(x,x')}{dx^2} + G(x,x') = \delta(x-x')$,

$$G(x=0) = G(x=\tfrac{\pi}{2}) = 0.$$

$y_1(x) = \sin x, \quad y_2(x) = \cos x.$

$$W(x) = \begin{vmatrix} y_1 & y_2 \\ y_1' & y_2' \end{vmatrix} = \begin{vmatrix} \sin x & \cos x \\ \cos x & -\sin x \end{vmatrix} = -1$$

$$\Rightarrow G(x,x') = \begin{cases} -\sin x \cos x', & 0 < x < x' \\ -\sin x' \cos x, & x' < x < \frac{\pi}{2}. \end{cases}$$

The solution of the inhomogeneous eqn. is

$$y(x) = \int_0^1 dx' \, G(x,x') f(x') =$$

$$-\cos x \int_0^x \sin\xi \, f(\xi)\, d\xi - \sin x \int_x^{\frac{\pi}{2}} \cos\xi \, f(\xi)\, d\xi.$$

5. The constrained Lagrangian is

$F(x, y, y') = (y')^2 + x^2 - \lambda y^2$, where

λ is the Lagrange multiplier.

The Euler-Lagrange eqn. is

$$\frac{d}{dx}\frac{\partial F}{\partial y'} - \frac{\partial F}{\partial y} = 0, \text{ or}$$

$y'' + \lambda y = 0$. Only $\lambda > 0$ is OK, so

$$y = a\cos\sqrt{\lambda}\,x + b\sin\sqrt{\lambda}\,x.$$

The boundary conditions yield $a = 0$

and $\sqrt{\lambda} = \pi n$, $n = 1, 2, 3, \ldots$

$\Rightarrow y(x) = b\sin(\pi n x)$. To find b, we use the constraint $\int_0^1 y^2\,dx = 2$,

which yields $b^2 \int_0^1 \sin^2(\pi n x)\,dx = 2$

$\Rightarrow b = \pm 2$, so

$y(x) = \pm 2\sin(\pi n x)$, $n = 1, 2, 3, \ldots$